POWER UP!

SUPPLYING ELECTRICITY

Most electricity is generated at sites such as power stations, wind farms, or nuclear power plants before being transmitted to homes and industries through a complex network known as a power grid. These power grids are made up of interconnected cables and other infrastructure, such as transmission towers and substations, that together deliver electricity over huge distances with as little power loss as possible. Electricity travels as a current of charged particles called electrons (see page 177).

ELECTRIC CURRENT

Electric current is the movement of tiny particles called electrons. In a metal wire, electrons move about freely but in no particular direction. When this wire is connected to a source of electrical power, such as a generator or a battery, the free electrons move in a single direction.